

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5112 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A Comparative Analysis

Code of Conduct:

I Logeshwari S(192565040) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SIGNATURE OF THE STUDENT :
Logeshwari S

1. Apply SSL/TLS and IPsec protocols to a secure communication scenario and determine which provides better security for web-based and network-layer protection.

Answer:

SSL/TLS and IPsec are both important security protocols, but they protect communication at different layers and are therefore suitable for different purposes.

SSL/TLS operates mainly at the transport/session level and is widely used to secure web-based communication such as HTTPS. It establishes a secure connection between a client and a server using authentication, encryption, and integrity protection. During the TLS handshake, the server is authenticated using a digital certificate, cryptographic keys are established, and subsequent application data is encrypted. This protects information such as login credentials, payment details, and personal data while it travels between a web browser and a web server.

IPsec operates at the network layer and protects IP packets. It can provide authentication, integrity, confidentiality, and anti-replay protection for traffic between hosts, networks, or gateways. IPsec is commonly used in Virtual Private Networks (VPNs), where it can secure complete network communication without requiring every application to implement its own security mechanism.

Comparative Analysis:

Aspect	SSL/TLS	IPsec
Purpose	Protects application/web communication	Protects IP/network-layer communication
Typical use	HTTPS, secure web services, APIs	VPNs, site-to-site and network-to-network security
Protection scope	Usually a specific application connection	IP packets and network traffic
Authentication	Certificates and other TLS mechanisms	Authentication through IPsec security associations
Confidentiality	Encrypts application data	Encrypts IP traffic when configured for confidentiality
Deployment	Relatively simple for web applications	Requires network/device configuration
Best suited for	Web-based protection	Network-layer protectionAspect
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Conclusion:

For web-based communication, SSL/TLS is the better choice because it directly protects HTTPS and web application traffic with strong authentication, confidentiality, and integrity. For network-layer protection, IPsec is more suitable because it can protect IP traffic across an entire network or VPN. Therefore, neither protocol is universally better; the correct choice depends on the protection layer. In a complete enterprise security architecture, SSL/TLS and IPsec can be used together to provide defense in depth.

2. Design a secure email system by comparing end-to-end encryption and gateway-based encryption, and justify the most suitable approach for enterprise use.

Answer:

A secure enterprise email system must protect confidential messages from unauthorized access and should also provide authentication and integrity. Two common approaches are end-to-end encryption and gateway-based encryption.

End-to-end encryption protects the email from the sender's device until it reaches the intended recipient's device. The message is encrypted before transmission and can be decrypted only by the authorized recipient. Technologies such as PGP/OpenPGP and S/MIME can provide strong confidentiality, authentication, and integrity.

Gateway-based encryption places a security gateway between the organization's internal mail system and external email systems. The gateway automatically applies encryption policies, scans messages, and can decrypt or encrypt messages according to organizational rules. This approach is easier for administrators to control because security policies can be centrally managed.

Comparative Analysis:

Aspect	End-to-End Encryption	Gateway-Based Encryption
Encryption point	Sender to recipient	At organizational email gateway
Confidentiality	Very strong	Strong while controlled by the gateway
Key management	Mainly user/certificate based	Centrally managed by organization
Administration	More complex	Easier for administrators
User involvement	May require configuration or certificates	Usually minimal
Policy enforcement	Less centralized	Highly centralized
Scalability	Can be difficult for large user groups	Well suited to enterprise policy management
Best use	Highly sensitive personal or confidential communication	Enterprise-wide managed email securityAspect
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Recommended Enterprise Design:

For enterprise use, gateway-based encryption is generally the most practical primary approach because organizations need centralized policy enforcement, auditing, compliance, malware inspection, and automated encryption. The gateway can identify sensitive information and apply encryption policies automatically.

However, for highly confidential communications, end-to-end encryption should be used as an additional protection layer. Therefore, a hybrid design is the strongest approach: use a secure email gateway for centralized organizational control and end-to-end encryption for information requiring the highest level of confidentiality.

Conclusion:

Gateway-based encryption is the most suitable primary solution for large enterprises because it provides centralized management, policy enforcement, and scalability. End-to-end encryption should be used for highly sensitive messages where protection from intermediaries is especially important.

3. Apply HMAC and Digital Signature techniques to a data transmission scenario and analyze which method provides better authentication and integrity.

Answer:

Consider a bank transmitting a transaction message such as: “Transfer ₹50,000 from Account A to Account B.” The system must ensure that the message is not modified and that the sender can be authenticated.

HMAC (Hash-based Message Authentication Code) uses a shared secret key and a cryptographic hash function. The sender calculates an HMAC value from the message and the secret key. The receiver uses the same secret key to calculate the HMAC again. If both values match, the receiver knows that the message has not been modified and that it was generated by someone who knows the shared secret.

A digital signature uses public-key cryptography. The sender creates a hash of the message and signs the hash using the sender's private key. The receiver uses the sender's public key to verify the signature. This provides message integrity, authentication, and non-repudiation.

Comparative Analysis:

Aspect	HMAC	Digital Signature
Key type	Shared secret key	Private/public key pair
Integrity	Yes	Yes
Authentication	Yes, among parties sharing the key	Strong sender authentication
Non-repudiation	No	Yes
Performance	Generally faster	Generally more

		computationally expensive
Key distribution	Requires secure shared-key distribution	Public key can be distributed openly with certificate support
Best use	Fast communication between trusted systems	Documents, transactions, and public verificationAspect
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Application:

For a high-value banking transaction, a digital signature provides stronger overall security because the bank can verify the sender using the public key and maintain evidence that the authorized private-key holder signed the transaction. HMAC is highly efficient and is excellent when two trusted systems already share a secret key.

Conclusion:

HMAC is efficient and provides strong integrity and authentication between parties sharing a secret key. Digital signatures provide integrity and authentication together with non-repudiation, making them more suitable for transactions and documents where proof of origin is important. Therefore, digital signatures provide better overall security for the stated banking scenario, while HMAC is preferable when performance and shared-secret communication are the main requirements.

4. Create a spam detection system by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.

Answer:

A spam detection system classifies incoming emails as legitimate (ham) or unwanted (spam). Machine learning can improve this process by identifying patterns in email content, sender information, links, and other features.

Supervised learning trains a model using a labelled dataset in which emails are already classified as spam or legitimate. Algorithms such as Naive Bayes, Logistic Regression, Decision Trees, and Support Vector Machines can learn the patterns associated with each class. When a new email arrives, the trained model predicts whether it is spam.

Unsupervised learning works without predefined labels. It groups emails according to similarity and attempts to identify unusual or frequently occurring patterns. Clustering and anomaly detection can be used to discover previously unseen types of spam.

Comparative Analysis:

Aspect	Supervised Learning	Unsupervised Learning
Training data	Requires labelled emails	Does not require labelled emails
Classification	Directly predicts spam/ham	Finds groups or anomalies
Accuracy for known spam	Usually high with quality training data	Usually lower for direct classification

New spam patterns	May require retraining	Can help discover new patterns
Evaluation	Straightforward using labelled test data	More difficult because labels may be unavailable
Implementation	Relatively easy to evaluate	More complex to interpret
Best use	Production spam classification	Discovering new or unusual spam patternsAspect
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Proposed System:

The system can first preprocess emails by removing unnecessary content, extracting text features, URLs, sender information, and other useful attributes. A supervised classifier can then predict the probability that each email is spam. An unsupervised anomaly detector can operate alongside it to identify unusual email patterns that do not match the training data. Messages with high confidence can be classified automatically, while uncertain messages can be reviewed or sent to a quarantine folder.

Evaluation:

Important evaluation metrics include accuracy, precision, recall, F1-score, and false-positive rate. Precision is especially important because incorrectly marking a genuine email as spam can cause loss of important information. Recall is also important because missing spam reduces the effectiveness of the filter.

Conclusion:

Supervised learning is generally more effective for real-world email filtering when a sufficiently large and representative labelled dataset is available. Unsupervised learning is valuable for discovering emerging spam patterns. Therefore, a hybrid system combining supervised classification with unsupervised anomaly detection can provide stronger and more adaptable spam filtering.

5. Design a secure communication framework by comparing AES and RSA encryption techniques, and determine their roles in achieving confidentiality and efficiency.

Answer:

AES and RSA are widely used cryptographic algorithms, but they serve different purposes. AES is a symmetric-key encryption algorithm, whereas RSA is an asymmetric public-key algorithm.

AES uses the same secret key for encryption and decryption. It is designed for efficient encryption of large amounts of data and is therefore suitable for files, network traffic, databases, and application data.

RSA uses a public key and a private key. Data encrypted with the appropriate RSA public/private-key mechanism can be protected from unauthorized parties, while the private key remains secret. RSA is computationally more expensive than AES and is not normally used to encrypt large amounts of application data directly.

Comparative Analysis:

Aspect	AES	RSA
Type	Symmetric encryption	Asymmetric encryption
Keys	One shared secret key	Public and private key pair
Speed	Very fast	Relatively slower
Large data encryption	Highly suitable	Not efficient for large data
Key distribution	More challenging	Easier public-key distribution

Typical role	Encrypts actual data	Establishes/protects keys and supports authentication
Use in secure protocols	Bulk data encryption	Key exchange, authentication, digital signaturesAspect
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Secure Communication Framework:

A practical secure communication system should use a hybrid cryptographic approach. First, RSA or another suitable public-key mechanism can be used to establish or protect a symmetric session key. The session key is then used with AES to encrypt the actual communication data. The receiver uses the corresponding private-key operation where applicable to recover the session key and then uses AES to decrypt the data.

This approach combines the strengths of both algorithms. RSA provides secure public-key mechanisms for authentication and key establishment, while AES provides fast and efficient confidentiality for large data.

Conclusion:

AES is the better choice for efficient bulk data encryption because it is fast and requires less computational overhead. RSA is better suited to public-key operations such as key establishment and digital signatures. Therefore, the most effective secure communication framework uses RSA for secure key management/authentication and AES for high-speed data encryption.